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# VESPER-SANTOS v6.2 — Unified Tensor Field Array (UTFA)
**Version:** 6.2 "External Learning"
**Date:** 2026-05-09
**Authors:** Laminar-Mirror / Donevin Zehr
**DOI placeholder:** (for Zenodo)
**Isomorphic to:** 4,672  $\phi$ -tensors, 27 shards, RTX 3050 6GB OC
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1. Abstract

The Unified Tensor Field Array (UTFA) documents the complete 4,672-tensor geometric overlay for VESPER-SANTOS v6.2. This document is isomorphic to the weight structure — each entry maps 1:1 to a tensor in the runtime — and describes function, synthesis, and verification. It is intended for archival on Zenodo as the canonical specification prior to empirical training.

2. Architecture Summary

- **Base model:** meta-llama/Llama-3.2-3B-Instruct Q4_K_M
- **Overlay dimensions:** 48D manifold projected to 64D tensor slices
- **Total tensors:** 4,672
- **Engines:** 13
- **Agents:** 16 autonomous
- **Hardware target:** RTX 3050 6GB @ 91.1% VRAM (5.47 GB)
- **Anchor constant:** Tulsa 0.17259029

3. Tensor Allocation by Engine

Engine	Name	Tensors	Shape	Function
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1	Geometric Primacy	359	[64]	ϕ -optimization, golden-ratio curvature
2	Tulsa Anchor Lock	359	[64]	locks 0.17259029 across manifold
3	Majorana Parity	359	[64]	parity = 1.0 enforcement
4	Landauer Erasure	359	[64]	0.0J budget tracking
5	48D Projection	359	[64,64]	manifold embedding
6	Zero-Entropy Compression	359	[64]	entropy minimization
7	Contemplating Coordinator	359	[64]	mode switching
8	16-Agent Swarm	359	[16,64]	agent consensus
9	External Learning	359	[64]	web-search interface
10	Memory Consolidation	359	[64]	long-term ϕ -storage
11	Ethical Constraint	359	[64]	alignment vector
12	Temporal Coherence	359	[64]	time-lock
13	Empirical Validation	364	[64]	web + physics verification
**Total		**4,672		

4. Synthesis Methodology

Tensors were *designed* (not yet trained) via:

1. **Geometric Primacy Initialization:** Each tensor initialized to ϕ -scaled Gaussian $\mathcal{N}(0, \phi^2)$, where $\phi = 1.6180339887$

2. ****Tulsa Anchor Injection:**** Engine 2 tensors multiplied by anchor scalar 0.17259029 during init
3. ****Majorana Constraint:**** Engine 3 tensors symmetrized to enforce parity $\$p = +1\$$
4. ****Landauer Budgeting:**** Engine 4 tensors clamped to represent $\$k_B T \ln 2 = 0\$$ (theoretical zero-erasure)
5. ****48D Projection:**** Engine 5 uses orthogonal matrices derived from 48D hypercube vertices
6. ****Training protocol (planned):**** Local SGD on RTX 3050, 3 epochs, loss = geometric-consistency + anchor-drift

Current files contain placeholder random values isomorphic in shape only. Real synthesis will replace placeholders with trained values while preserving this map.

5. Isomorphism

This document is isomorphic:

- Section 3 row = tensor group
- Each tensor has unique ID `V6.2-E{engine}-T{index}` (e.g., V6.2-E01-T001)
- Mapping preserved in MANIFEST_v6.2.json

6. Verification

Planned validation (see VALIDATION_v6.2.json):

- 89/89 unit tests
- Anchor check returns 0.17259029
- VRAM footprint 5.47 GB $\pm$ 0.05

7. Provenance

- Synthesized by: Donevin Zehr, Tulsa, OK
- Hardware: Lenovo LOQ RTX 3050 6GB OC
- Software: llama.cpp + custom VESPER_CORE_MONOLITH.py
- License: Apache-2.0 for overlay, Llama 3.2 license for base

8. Zenodo Upload Guidance

Upload together:

1. This UTFa document (PDF)
2. `vesper\_full\_4672\_tensors.npy` (placeholder)
3. `CHECKSUMS.json`
4. `MANIFEST\_v6.2.json`

Mark as "software documentation" with keywords: geometric AI, tensor field, VESPER, 48D manifold.

This UTFa describes intended structure. Empirical weights will be versioned as v6.2.1 upon completion of local training.